## Radian measure

**1**

**a** $\frac{\pi}{180}$

**b** $\frac{180}{\pi}$

**2**

**a** $2\pi$ radians

**b** $\pi$ radians

**c** $\frac{\pi}{2}$ radians

**d** $\frac{5\pi}{4}$ radians

**e** $-\frac{5\pi}{3}$ radians

**f** $\frac{28\pi}{45}$ radians

**g** $\frac{29\pi}{180}$ radians

**h** $2.81$ radians

**3**

**a** $60^\circ$

**b** $30^\circ$

**c** $45^\circ$

**d** $270^\circ$

**e** $225^\circ$

**f** $-120^\circ$

**g** $-300^\circ$

**h** $240.6^\circ$

**4**

**a** True

**b** False. $60^\circ$ must equal to $\frac{\pi}{3}$ radians, because it is $\frac{1}{3}$ of $180^\circ$.

**c** False. $30^\circ$ must equal to $\frac{\pi}{6}$ radians, because it is half of $60^\circ$.

**d** True

**e** True

**f** False. $225^\circ$ must be equal to $\frac{5\pi}{4}$ radians, because it is five groups of $45^\circ$.

## Coterminal angles

**5**

**a** $\frac{21\pi}{8}$

**b** $-\frac{11\pi}{8}$

**6**

**a** $\frac{25\pi}{8}$

**b** $-\frac{7\pi}{8}$

**7**

**a** $\frac{4\pi}{3}$

**b** $-\frac{8\pi}{3}$

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**8**
**a** Quadrant 3 **b** $\frac{\pi}{3}$ **c** $\frac{\pi}{3}$

**9**
**a** Quadrant 3 **b** $\frac{\pi}{7}$ **c** $\frac{\pi}{7}$

**10**
**a** $454^\circ$ **b** $-266^\circ$ **c** Quadrant 2

**11**
**a** $302^\circ$ **b** $-418^\circ$ **c** Quadrant 4

**12**
**a** $129^\circ$ **b** $29^\circ$

**13**
**a** $\frac{\pi}{4}$ **b** $\frac{\pi}{3}$

**14**
**a** $360n$ **b** $360n$

**15**
**a** $2\pi n$ **b** $2\pi n$

## Reference angles

**16**
**a** $69^\circ$ **b** $17^\circ$ **c** $85^\circ$

**17**
**a**
**i** $193^\circ$ **ii** $13^\circ$
**b**
**i** $159^\circ$ **ii** $21^\circ$

**18** $30^\circ$

**19** $65^\circ$

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**20** $30^\circ$

**21**

**a** $\frac{14\pi}{13}$

**b** $-\frac{12\pi}{13}$

**22**

**a** $\frac{6\pi}{7}$

**b** $-\frac{8\pi}{7}$

**23**

**a** $\frac{21\pi}{11}$

**b** $-\frac{\pi}{11}$

**24** $\frac{\pi}{11}$

**25** $\frac{\pi}{7}$

**26** $\frac{\pi}{5}$

**27** $\frac{\pi}{3}$ radians

**28**

**a** $\frac{10\pi}{9}$ radians

**b** $\frac{\pi}{9}$ radians

**29**

**a** $\theta = \frac{\pi}{17}, \frac{16\pi}{17}$

**b** $\theta = \frac{18\pi}{17}, \frac{33\pi}{17}$

**c** $\theta = \frac{\pi}{17}, \frac{33\pi}{17}$

**d** $\theta = \frac{18\pi}{17}, \frac{16\pi}{17}$

**e** $\theta = \frac{\pi}{17}, \frac{18\pi}{17}$

**f** $\theta = \frac{16\pi}{17}, \frac{33\pi}{17}$

**30**

**a** $\frac{4\pi}{13}$

**b**

**i** $\alpha = \frac{9\pi}{13}, \frac{4\pi}{13}$

**ii** $\alpha = \frac{9\pi}{13}, \frac{17\pi}{13}$

**iii** $\alpha = \frac{9\pi}{13}, \frac{22\pi}{13}$

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**31**

a $\frac{\pi}{17}$

b

i $\alpha = \frac{18\pi}{17}, \frac{33\pi}{17}$

ii $\alpha = \frac{18\pi}{17}, \frac{16\pi}{17}$

iii $\alpha = \frac{18\pi}{17}, \frac{\pi}{17}$

**32**

a $\frac{2\pi}{17}$

b

i $\alpha = \frac{32\pi}{17}, \frac{19\pi}{17}$

ii $\alpha = \frac{32\pi}{17}, \frac{2\pi}{17}$

iii $\alpha = \frac{32\pi}{17}, \frac{15\pi}{17}$

**33**

a $\frac{8\pi}{17}$

b

i $\theta = \frac{25\pi}{17}, \frac{26\pi}{17}$

ii $\theta = \frac{8\pi}{17}, \frac{26\pi}{17}$

iii $\theta = \frac{8\pi}{17}, \frac{25\pi}{17}$

**34**

a $\frac{2\pi}{17}$

b

i $\theta = \frac{2\pi}{17}, \frac{15\pi}{17}$

ii $\theta = \frac{2\pi}{17}, \frac{32\pi}{17}$

iii $\theta = \frac{15\pi}{17}, \frac{32\pi}{17}$

**35**

a $\frac{5\pi}{13}$

b

i $\theta = \frac{5\pi}{13}, \frac{8\pi}{13}$

ii $\theta = \frac{8\pi}{13}, \frac{18\pi}{13}$

iii $\theta = \frac{5\pi}{13}, \frac{18\pi}{13}$

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